The following listing shows a simple implementation of such a queue based on a cut-down version of the interface to the queue in listing 6.2; you have only one `try_pop()` function and no `wait_and_pop()` because this queue supports only single-threaded use.

Listing 6.4. A simple single-threaded queue implementation

```
template<typename T>
class queue
{
private:
    struct node
    {
        T data;
        std::unique_ptr<node> next;

        node(T data_):
            data(std::move(data_))
        {}
    };

    std::unique_ptr<node> head;    <-(1)
    node* tail;      <-(2)
public:
    queue()
    {}
    queue(const queue& other)=delete;
    queue& operator=(const queue& other)=delete;

    std::shared_ptr<T> try_pop()
    {
        if(!head)
        {
            return std::shared_ptr<T>();
        }
```